

SHORTEST PATHS IN A RACE

FROM CALC MADE EASY PAGE 300

WHEN RACING AROUND A SHARP CORNER, WE HAVE THE OPTION TO CUT ACROSS ROUGH TERRAIN, AT A REDUCED SPEED. THE CORNER IS **1** UNIT AWAY, THE DESTINATION IS **b** UNITS FROM THE CORNER. IF WE TAKE A SHORTCUT **x** UNITS FROM THE CORNER, WE ONLY HAVE **h** UNITS LEFT, BUT AT A FRACTION **γ** OF OUR NORMAL SPEED. THE TOTAL TIME IS THUS

$$t = \underbrace{(1-x)}_{\text{TIME BEFORE WE CUT}} + \underbrace{h/\gamma}_{\text{SHORTCUT TIME}}$$

DIFFERENTIATING,

$$\frac{dt}{dx} = \frac{x}{\gamma(x^2+b^2)^{1/2}} - 1$$

WHICH IS MINIMIZED AT

$$\hat{x} = b \sqrt{\frac{\gamma^2}{1-\gamma^2}}$$

THE BEST SHORTCUT IS THUS A FUNCTION OF $b \pm \gamma$. FOR ANY SPECIFIC b , WE CAN ADJUST γ TO MAKE $\hat{x} \rightarrow 0$ AS $\gamma \rightarrow 0$, SO NEVER CUT, TO $\hat{x} = 1$ IF $\gamma \geq (1+b^2)^{-1/2}$, SO IMMEDIATELY CUT.

IN *CALC MADE EASY*, $b = 3/4$, $\gamma = 3/4$, SO $\hat{x} = 0.85$, ABOUT 15% ALONG (MARKED IN FIG. 3)

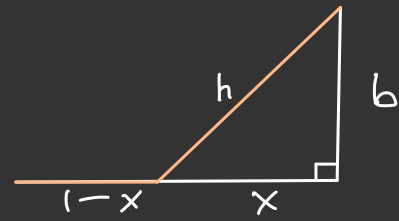


Fig. 1

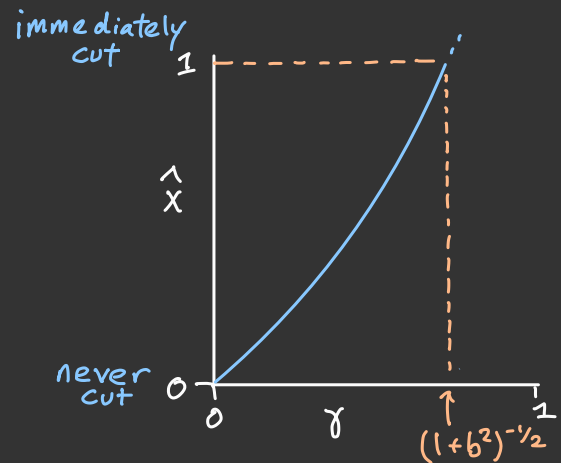


Fig. 2

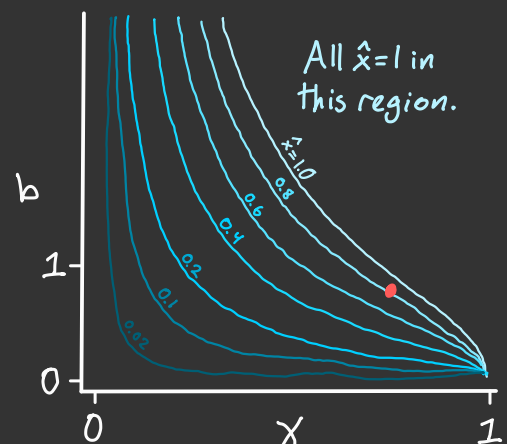


Fig. 3